

WORKING PAPER · NO. 2025-56

Large Language Models, Small Labor Market Effects

Anders Humlum and Emilie Vestergaard

MAY 2025 (UPDATED SEPTEMBER 2025)

Large Language Models, Small Labor Market Effects*

Anders Humlum[†]

Emilie Vestergaard[‡]

September 22, 2025

Abstract

We examine the early labor market impacts of AI chatbots by linking large-scale, representative adoption surveys to administrative labor records in Denmark. Using difference-in-differences, we estimate precise null effects on earnings and recorded hours at both the worker and workplace levels, ruling out effects larger than 2% two years after. These null results hold for intensive users, early adopters, workplaces with substantial investments, workers reporting large gains, flexible-pay occupations, and early-career jobs. Adoption is linked to occupational switching and task restructuring, but without net changes in hours or earnings. Our findings challenge narratives of imminent disruption from Generative AI.

*We thank David Autor, Marianne Bertrand, Jacob Conway, Matthew Notowidigdo, Lindsey Raymond, Chad Syverson, and seminar participants at the 2025 NBER SI: Digital Economics and Artificial Intelligence, the 2025 SED Meeting: Automation, the Becker Friedman Institute, Chicago Booth, the Chicago Fed, ETUI, INSEAD, Microsoft Research, and MIT Sloan FutureTech for helpful comments and suggestions. This project received financial support from the Center for Applied Artificial Intelligence and the Polsky Center for Entrepreneurship and Innovation. Caspar Ringhof provided excellent research assistance.

[†]University of Chicago, Booth School of Business, and NBER, anders.humlum@chicagobooth.edu

[‡]University of Copenhagen, Department of Economics, emilievestergeraad@econ.ku.dk

to labor market outcomes.

To examine these outcomes, we link our surveys to administrative records on monthly earnings, hours worked, and occupations through December 2024—two years after the launch of ChatGPT. Using a difference-in-differences framework, we compare adopters and non-adopters before and after the arrival of AI chatbots, estimating impacts at both the worker and workplace levels.

We begin by analyzing workers who use AI chatbots. If these tools enhance individual productivity, a natural question is whether such gains translate into higher earnings. Our first key finding is that AI chatbots have had minimal overall impact on workers who use them. Difference-in-differences estimates for earnings, recorded hours, and wages are all precisely estimated zeros, with confidence intervals ruling out average effects larger than 2%. These null results hold even for workers who use the tools daily, report substantial productivity gains, perform new AI-related tasks, and work in firms that encourage and invest in chatbots. Moreover, we find no significant effects in any of our 11 occupations, including those with flexible, decentralized wage-setting.

While chatbots have not led to earnings gains for users, we find a clear link to occupational mobility: adopters are more likely to have transitioned into new occupations after the arrival of AI chatbots. This aligns with experimental evidence that chatbots are especially beneficial for workers with less prior expertise (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023), and supports the view that Generative AI helps workers extend into new occupations (Autor, 2024). However, the effects are modest—amounting to 4% of full-time equivalent employment—and do not result in net gains in hours or earnings.

Next, we zoom out to the workplace level and examine whether employers that encourage and invest in AI chatbots have experienced differential changes in labor demand. If chatbots substitute for labor, they could lead to slower hiring or trigger layoffs; if they enhance productivity, firms might expand hiring. Yet we find no such differences: workplaces that encourage chatbot use show no differential changes in employment or

where we add adopter-specific time trends to ensure our pooled difference-in-differences estimates are not driven by spurious trends in outcomes. The parameters of interest, β_y , measure how labor market outcomes for adopters changed in year y after the introduction of AI chatbots, compared to the period before their arrival. We estimate Equation (2) by OLS and cluster the standard errors by workplace-occupation level.

3 Results

3.1 Worker Earnings

Figure 2, Panels (a) and (b) examine earnings of chatbot adopters, distinguishing between those with and without employer encouragement. Each group is compared to non-encouraged non-adopters—workers who are most insulated from AI chatbots, either through their own non-use or that of their coworkers.

Panel (a) begins by presenting differences in means, controlling for workers’ pre-determined characteristics—occupation, gender, age, and experience. Adopters appear to earn substantial premia of about 9% for those with employer-encouraged usage and 4% for those without. However, leveraging the panel dimension of our administrative data reveals that these earnings differentials entirely predate the arrival of AI chatbots. When indexing differences to November 2022, the difference-in-differences estimates in Panel (b) show no differential changes for adopters following the introduction of AI chatbots. The estimates remain flat even among adopters whose employers explicitly encourage chatbot use. Appendix E.2.1 shows that these null results hold separately for workers’ hourly wages, intensive-margin hours, and extensive-margin employment.¹⁰

Figure 2, Panel (b) provides three key insights. First, two years after their launch, AI chatbots have had minimal earnings impact on workers who use them. The confidence intervals of our dynamic difference-in-differences estimates rule out changes larger than

¹⁰The results for adopter earnings echo the pattern documented by Krueger (1993) and critiqued by DiNardo and Pischke (1997): although chatbot use is correlated with sizable cross-sectional earnings premia, these differentials vanish once accounting for positive selection.

3%, and our pooled difference-in-differences estimates rule out effects larger than 2%.¹¹ Second, these effects show no differential trends after the introduction of AI chatbots, indicating that the minimal impacts are not merely a very short-term phenomenon. Lastly, despite evidence in Section 1.4 that employer encouragement boosts work-related benefits of AI chatbots, we find no differential changes in earnings for workers whose employers encourage their chatbot use.

Heterogeneity by Strength of Treatment. Figure 4, Panel (a) examines the earnings impacts of chatbot use by the strength of the adoption treatment.

The top three rows consider the effects of additional investments in employer-provided chatbots and training. These estimates offer insights into the impacts of AI chatbots when firms pursue a more comprehensive strategy. Yet even when the “full package” of encouraged use, enterprise chatbots, and training substantially increases reported benefits (Section 1.4), we find no detectable impact on adopter earnings.

The next two rows focus on more intensive users—those who engage with AI chatbots daily or who adopted early (within the first year after ChatGPT’s launch). These workers have not experienced earnings gains either.

The remaining rows examine workers who report greater benefits from AI chatbot usage: time savings exceeding one hour per day, higher-quality output, enhanced creativity, and engagement with new AI-related tasks. While these self-reported benefits are substantial, the estimates show none of them is associated with better labor market outcomes in our administrative data.

Taken together, Figure 4.(a) suggests that the null earnings effects are not simply the result of weak treatments—at either the firm or worker level. Workers in firms that adopt a more active, multifaceted approach to AI chatbots do not see differential earnings impacts. Nor do those who use the tools more frequently, adopted them earlier, or report

¹¹The dynamic difference-in-differences in Figure 2.(b) reveal that adopters are on slightly stronger labor market trends. However, because these trends entirely predate AI chatbots, the pooled difference-in-differences in Figure 5.(a) (which control for pre-trends; see Equation (2)) are precise zeros, with confidence bands ruling out effects larger than 2%.